Fiscal Year:	FY 2024	**Task Last Updated:**	FY 04/30/2024
PI Name:	Lewis, Norman G Ph.D.		
Project Title:	An Integrated Omics Guided Approach to Lignification and Gravitational Responses: The Final Frontier		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	05/01/2015	**End Date:**	04/30/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 4/30/2023 per NSSC information (Ed., 5/4/22) NOTE: End date change to 4/30/2022 per NSSC information (Ed., 4/2/21) NOTE: End date change to 4/30/2021 per NSSC information (Ed., 3/10/2020) NOTE: End date changed to 4/30/2020 per PI (Ed., 3/4/19) NOTE: End date changed to 4/30/2019 per NSSC information (Ed., 8/16/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Davin, Laurence Ph.D. (Washington State University) Hanson, David Ph.D. (University of New Mexico) Lipton, Mary Ph.D. (Battelle Memorial Institute) Sayre, Richard Ph.D. (New Mexico Consortium) Starkenburg, Shawn Ph.D. (Los Alamos National Security)		
Grant/Contract No.:	NNX15AG56G		
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Task Description:	We propose a multi-omics study using the model plant Arabidopsis under both 1g and in microgravity conditions (spaceflight). Our approach spans comprehensive phenomics, metabolomics, transcriptomics, and proteomics strategies, and is incisively and uniquely melded via deployment of an integrated computational biology (ICB) approach we are pioneering. Our plant lines include wild type, various mutants we have generated with different lignin amounts through manipulation of the multigene family encoding arogenate dehydratases, and other lines enhanced in carbon assimilation capacity, and combinations thereof. We consider this places us in an unprecedented position to investigate how plants function in altered gravity environments. We are very well positioned for incisive spaceflight and definition stage (1g) studies to investigate gene/metabolic network relationships and adaptations resulting from varying lignin and carbon assimilation levels, e.g., on photosynthesis; C allocation; water use efficiency (WUE); vascular plant growth/development; vasculature performance; auxin transport; and gravitational adaptations. Our overarching hypothesis is that a comprehensive interrogation (an integrative omics study) of our Arabidopsis lines with varying lignin levels and/or modulated carbon concentrating mechanisms (CCMs) or combination thereof will identify gene/metabolic networks, mechanisms and/or pathways that are differentially modulated at 1g and on exposure to microgravity, i.e., various omics (phenomics, transcriptomics, genomics, proteomics, metabolomics, and ICB) will allow us to study these in a truly unprecedented way. Overall objectives: 1. Establish multi “omics” effects of modulating lignin and CCM levels i) at 1g and ii) in spaceflight. 2. Compare/contrast data, using an ICB approach, to better define and understand gravity sensing and responses, and if threshold/induction parameters are modified or changed, when lignin and CCM levels are varied. More specifically, we address distinct hypotheses for our various teams, and integrate, dissect, and incisively analyze them holistically in a manner hitherto not possible. These 5 hypotheses include that: modulating lignin and CCM levels differentially affect carbon assimilation/re-allocation, photosynthesis, and WUE (Team 1); modulating lignin and CCM levels differentially affect secondary and primary metabolite levels (metabolomics) (Team 2); system-wide modification in the transcriptome occurs through a common transcriptional regulatory mechanism, and transcriptome/proteome “discrepancies” result from over-simplification of transcript analyses (Team 3); differential alterations in lignin and CCM levels can often be attributed to overall distinct changes in protein expression and phosphorylation patterns in a defined set of proteins (Team 4); an integrated omics analysis will provide urgently needed new insights into global effects on plant biological processes at both 1g and in microgravity (Teams 1-4). Each hypothesis draws upon the most advanced technologies available for study. We consider that our ICB approach will transform omics analysis through our advanced instrumentation and analytical tools. We will utilize (or design) computational tools/mathematical algorithms for integration and correlation of high resolution phenotype measurements (phenomics) with 'low' resolution global subcellular system measurements (transcriptomics, etc.) through 'nth' dimensional analysis. Our study aligns with Research Emphasis 1 and 3, and decadal survey elements in Cell, Microbial, and Molecular Biology (CMM-3, CMM-5), Organismal and Comparative Biology (OCB 2-5), Developmental Biology (DEV-4), and Plant and Microbial Biology, chapter 4 (P2). Our data generation will also be seamlessly integrated with various web-based platforms to handle, disseminate, and inter-actively utilize through iPlant and OpenMSI, and thus are made available to NASA as well as being a community resource.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	This research will provide NASA and science in general with the first “big ‘omics’ data” analysis, integration, and assessment – at the gene, protein, and metabolic outcome levels – as to how microgravity alters the basic responses of plants when the influence of gravity is removed/minimized. This will serve as the foundation “omics” analyses in subsequent spaceflight and colonization experiments in space, as well as shedding new insights into the manifold effects of gravity during plant growth and development.		
	Overview: Comprehensive ‘multi-omics’ analyses of model plant species, thale cress (Arabidopsis thaliana), grown for up to circa 6 weeks duration after germination on International Space Station (ISS) with corresponding Kennedy Space Center (KSC) ground controls. [Multi-omics analyses refer to metabolomics, proteomics, transcriptomics, phenomics, and (to the extent possible) anatomical analysis of lignified stems. A glossary of terms is at end of summary.] Biological Partners: Six Arabidopsis genotypes, wild type (WT) Columbia, Columbia modified ostensibly to increase plant biomass, and four lignin-reduced Columbia genotypes. Lignin provides plant cell wall reinforcement, thereby enabling plants to stand upright. Lignin-reduced genotypes, unable to stand upright, were utilized to determine whether structural lignin reinforcement was required in microgravity on ISS. These genotypes offered the exciting opportunity to investigate the central question as to whether there was a threshold lignin level needed for Arabidopsis to stand upright in microgravity. Consortium Team Partners: This included research scientists from Washington State University (WSU), University of New Mexico (UNM), Los Alamos National Laboratory (LANL), Pacific Northwest National Laboratory (PNNL), Kennedy Space Center (KSC), and ISS astronauts. Objectives and results: 1). Establish “multi-omics’ effects of modulating lignin and carbon concentrating mechanism (CCM) levels at 1g and in spaceflight on ISS; 2). Compare/contrast data, using an integrated computational biology (ICB) approach, to better define and understand gravity sensing and responses and if threshold/induction parameters are modified or changed when lignin and CCM levels were varied. Pre-ISS consortium tasks successfully completed 2015-2018 in conjunction with KSC Payload Development Team (PDT): Generated CCM genotype with small, but measurable, biomass increases and increased stem fiber cell wall thickness, whereas the lignin-reduced adt mutants had weakened vascular bundle cell walls; Established APH Science Carrier Arabidopsis growth conditions for KSC Science Verification Test (SVT), Experiment Verification Test (EVT), and ISS; Demonstrated Arabidopsis seed viability for prolonged storage (24 weeks) in Science Carriers; Identified transportation conditions to ISS for Arabidopsis seeds in Science Carrier, without seeds being dislodged during transit; Scientific/technical evaluation at KSC of suitability of hand-held photosynthesis measurement (FluorPen) device; Quality control criteria established for Arabidopsis tissue sample preparations for multi-omics analyses; Two multi-omics foundational analyses conducted using soil-grown Arabidopsis genotypes, as baseline studies to compare with ISS/KSC investigations. Soil-grown Arabidopsis lignin deficient genotypes had reductions in related secondary metabolite levels (flavonoid, phenylpropanoid, glucosinolate, and other metabolite classes). By contrast, transcriptomic and proteomic analyses of corresponding biochemical pathway steps generally showed increased abundances, but such responses were ultimately futile in restoring either lignin or secondary metabolite levels. ISS prototype APH flight unit installation and functional verification: Astronaut Ricky Arnold completed on-orbit Advanced Plant Habitat (APH) installation, with first power-up and 5-day functional checkout on December 01, 2017. Science Carrier with WT Arabidopsis and dwarf wheat seeds in the ISS APH prototype were then successfully flood filled. Plants seemingly looked healthy over a 37-day growth period in Spring 2018. ISS and KSC Arabidopsis genotype sequential grow-outs: This was the inaugural investigation for the ISS APH prototype, and the project was named PH-01 by NASA personnel. 192 Arabidopsis rosette leaf and stem tissues were collected for multi-omics analyses at circa 6 week growth/development/lignification. Roots were not harvested as APH Science Carrier technical design precluded this. PH-01 Grow-out 1: 46 Arabidopsis plants (Day 45) harvested on ISS stored in –160 °C GLACIER freezer until return to Earth (Splashdown, Pacific Ocean, January 13, 2019), with frozen Arabidopsis samples transported and delivered to Institute of Biological Chemistry (IBC), Washington State University (WSU). Samples were immediately stored at –80 °C at WSU until needed. KSC Ground Control Arabidopsis tissues (from 46 plants) were transported/delivered in frozen form to IBC/WSU and stored as above.		

Task Progress:

PH-01 Grow-out 2: PH-01 Grow-out 2 was highlighted in NASA's first 8K video (1.2 M views to date since its release 5 years ago; https:// 43 similarly frozen plants, following harvesting on Day 45, were returned to Earth January 7, 2020, then transported/delivered frozen to IBC/WSU and stored as above. KSC Ground Control Arabidopsis tissues (from 40 plants) were also transported/delivered in frozen form to IBC/WSU and stored as above.

APH plant health, plant genotype validation, and APH functional performance assessments: Overhead and side camera images were used daily to assess plant health and growth, and stem orientations, during ISS and KSC Grow-outs. Plant growth was estimated by measuring rosette leaf areas, where it was concluded that ISS-grown plants lagged in growth/development relative to KSC counterparts. PAM photosynthesis measurements showed higher variability on ISS, relative to KSC, presumably reflecting higher stresses experienced on ISS. Both ISS and KSC grown Arabidopsis genotypes, nevertheless, all looked robust and healthy visually. Stems of ISS lignin-reduced mutants were, however, unable to stand upright, and subsequently discovered have to abnormal stem vasculature cross-sections when compared to KSC grown specimens – all consistent with a higher lignin level threshold being needed to grow upright, even in microgravity. This was an unexpected finding, and opposite to conventional wisdom.

Arabidopsis tissues, for multi-omics analyses, had genotypes validated on return to Earth. Of the 192 specimens evaluated by PCR, 4 rosette leaf samples did not have the genotypes as assigned to their designated APH Science Carrier position number, and these were accordingly amended.

ISS and KSC ICB multi-omics analyses: The comprehensive analyses and 'big data' compilation represented a significant spaceflight (ISS) multi-omics ICB baseline study for Arabidopsis. However, meaningful integration of multi-omics data is a well-known problematic issue across all biology disciplines. Accordingly, our inter-disciplinary team effort focused upon integration and analysis of transcriptomics, proteomics, and metabolomics (including initial emphasis on secondary and primary metabolites, lipids, and lignin). Our approach was designed to tease out key ISS spaceflight environment effects relative to KSC counterparts.

ICB metabolomics, proteomics, and transcriptomics analyses: ISS lignin levels were generally lower than KSC Ground Controls and resulting from slower plant growth on ISS. Secondary metabolite (stem and rosette leaf) profiles for both ISS and KSC Grow-outs had similar profiles, with flavonoid, phenylpropanoid, glucosinolate, and shikimate-chorismate pathway levels generally becoming (increasingly) lower in lignin-deficient genotypes. The mainly primary metabolite analyses of stems established shikimic acid and phenylpropanoid levels on average also decreased in genotypes most lignin-deficient, relative to WT, for both stem/rosette leaf tissues in ISS/KSC Grow-outs 1 and 2. Aromatic amino acids, Phe and Tyr, in stems showed similar trends whereas in rosette leaves results were more varied for the same genotypes. In contrast, bona fide biochemical steps for the above secondary metabolite pathways generally showed increased proteome and transcriptome abundances in lignin-deficient genotypes, but these were again futile in restoring either lignin or secondary metabolite WT genotype levels, whether ISS or KSC grown.

Towards determining ICB transcriptomic and proteomic effects on photosynthesis, carbon fixation, and photorespiration: Our ICB approach also targeted photosynthesis, carbon fixation, and oxidative phosphorylation to evaluate whether photosynthesis and photorespiration were affected. Transcription profiles from Grow-out 2 (with transcripts from chloroplast and mitochondrial encoding proteins retained) indicated that Photosystems I and II subunits were down-regulated in a lignin-reduced genotype (ISS vs KSC), as were two rate-limiting Calvin cycle enzymes. Our preliminary analysis of global transcript levels clearly revealed a dramatic shift in transcriptome machinery and Calvin cycle proteins, but it remains to be determined whether one or more mechanisms are responsible for the varied genotype responses to differing lignin contents. Proteomic analyses were apparently similar, together with indications that photorespiration had increased in lignin-deficient genotypes.

Towards better understanding ICB transcriptomic and proteomic effects for radiation and DNA repair on ISS-grown Arabidopsis genotypes: Numerous genes associated with base/nucleotide excision repair and homologous recombination were statistically differentially expressed (DE) in the ISS spaceflight microgravity environment. The largest lignin-deficient mutants had more DE, suggesting different genotype responses to increased radiation in the ISS spaceflight environment. Proteins related to radiation damage repair in ISS Grow-out 1 also increased in abundance, as compared to KSC counterparts. DNA repair and recombination proteins, DNA replication, and homologous recombination proteins were additionally affected. This observation may suggest that plants with reduced lignin levels were more prone to DNA damage in the ISS spaceflight environment.

Towards determining ICB single nucleotide polymorphism (SNP) effects on ISS grown Arabidopsis genotypes: Our transcription analysis unexpectedly suggest that mutation rates might be lower in ISS vs KSC samples. This was unexpected given the ISS spaceflight environment (including radiation and microgravity) experienced. A provisional hypothesis for these results is possibility of over-expression of DNA repair mechanisms resulting from exposure to the ISS spaceflight environment. This would be consistent with the observed differential regulation of DNA repair mechanisms observed. Such over-expression could cause more mutations to be corrected, as compared to the baseline rate of mutations observed with KSC samples. Alternatively, ISS plants were possibly more stressed due to microgravity and increased radiation, thereby (at least in part) causing slower organismal growth rates. Slower growth rates would correspond to slower and fewer cell divisions, which could conceivably reduce the number of observed mutations across all cells in each tissue sample.

Towards correlating ICB proteomic and transcriptomic ISS spaceflight responses as regards cell signaling and microbial perception/susceptibility for ISS Arabidopsis genotypes: Initial analysis of differential expression observed in ISS (as compared to KSC) samples revealed more differential regulation in leaf compared to stem tissues. This trend was present in both transcriptomics and proteomics datasets, albeit to a lesser extent with protein abundance differences. Functional Term enrichment analysis showed photosynthesis-related proteins significantly down-regulated in leaf samples on ISS compared to KSC, this being consistent with previous studies of plant responses to stress conditions. Up-regulation at the transcript level for leaf samples on ISS was concentrated in signaling, and related genes, as well as genes involved in systemic acquired resistance (SAR). Various proteins related to cell signaling followed the same trend and increased in abundance in Arabidopsis plants with least lignin contents. For stem samples on ISS, no GO terms were significantly enriched among protein sets, but for transcriptomic features, down-regulation of growth and development and up-regulation of defense responses and signaling highlighted the list of regulatory differences. Several disease resistant proteins, and precursors to a hypersensitive response (HR), were significantly upregulated, perhaps suggesting attempted acclimation to the ISS spaceflight microgravity environment. Exosome and membrane trafficking proteins are often involved in protection against pathogenic attack. As lignification also plays a role in plant-pathogen/microbial defenses, it may be that ISS grown plants with reduced lignin levels are more prone to increased pathogen susceptibility in the spaceflight ISS microgravity environment.

Towards completion of IBC multi-omics analyses and visualization: Significant progress was made in integrating/comparing the diverse multi-omics datasets to gain both new and comprehensive insights into the systems-level response(s) to the ISS spaceflight microgravity environment with the six Arabidopsis genotypes. Metabolomics data was used as a cross-check. Pathway analysis used targeted and untargeted approaches. The former involved extracting features from each dataset associated with various structural (biochemical) pathways and comparing fold changes in ISS vs KSC across genotypes. The latter involved use of a LANL developed tool OPaver (Omics Pathway Viewer). Opaver enabled us to map differential expression/abundance values onto curated KEGG pathway maps for multi-omics datasets. Our analysis to date indicates that some metabolic pathways were augmented by the ISS spaceflight environment conditions vs KSC, but analysis is still ongoing.

Outreach:

Outreach was an important component of this ISS/KSC study, but with no direct support from the NASA supported project here. The Outreach components are summarized further below.

Summary and future steps: An important and continuing contribution was in providing the best possible multi-omics integration analyses for the various Arabidopsis genotypes grown on ISS and at KSC. Our massive datasets and other documentation, although embargoed until publication, are currently being provided to NASA Gene Lab for curation, as well as to data repositories required for publication. It is anticipated that our multi-omics integration will continue to take an additional 12 months or so to fully complete – including giving additional attention to effects of radiation on ISS and potentially different effects on the various Arabidopsis genotypes. Ongoing analyses include determining the full extent of physiological modulations induced in the ISS spaceflight microgravity environment, by: 1) more detailed mapping of transcripts to specific intra-cellular compartments (i.e. chloroplast, mitochondria, etc.) in order to more fully understand how microgravity on ISS modulates various metabolic pathways; 2) deducing what trends may be uniquely attributed to the ISS spaceflight microgravity environment; 3) further addressing the proposal hypotheses that system-wide modifications noted in ADT KO and WT Arabidopsis genotypes may be caused (in part) through a common transcriptional regulatory mechanism.

Glossary: Metabolomics here included analyses and quantification of secondary metabolites (many with plant defense functions), lipids (largely fatty, waxy, or oily compounds), primary metabolites (considered essential for plant growth and development), and lignin. Transcriptomics and proteomics were used for detection of gene expression and corresponding protein/enzyme profiles (and their individual abundances) at the circa 6 week harvesting stage for both ISS and KSC Grow-outs. Phenomics included photosynthesis measurements using a hand-held pulse-amplitude monitoring (PAM) device, used for the first time on ISS, as well as in monitoring of rosette leaf tissue areas and of 'bolting' stem orientations during Arabidopsis growth/development/lignification. Cell wall anatomical analyses of both ISS and KSC Arabidopsis stems were used to determine whether ISS grown Arabidopsis stem vasculature differed from corresponding KSC Ground Controls.

Unexpected setbacks: Germination and plant death/dying difficulties were experienced in both ISS Grow-outs. This required analysis of ISS APH functionality data records, as such difficulties did not occur during KSC Grow-outs. It was concluded that there was uneven APH watering in growing all needed plant genotypes and, in particular, for ISS Grow-out 1, as well as intermittent ISS APH partial lighting failures for Grow-out 2. Furthermore, analysis of water availability at different growth positions in the APH on ISS was a complication that can hopefully be better teased out from further analyses of multi-omics and ISS APH data for both Grow-outs. In both cases, together with slower growth on ISS, only one 6 week harvest was possible for multi-omics analyses.

Supplementary (Publications to Date, Newsletters, Seminars/Lectures, Presentations, Significant Media Coverage, Training, and Outreach)

Publications:

Höhner, R., Marques, J.V., Ito, T., Amakura, Y., Budgeon, A.D., Jr., Weitz, K., Hixson, K.K., Davin, L.B., Kirchhoff, H., and Lewis, N.G. 2018. Reduced arogenate dehydratase expression: Ramifications for photosynthesis and metabolism. Plant Physiol. 177: 115-131. http://dx.doi.org/

Hixson, K.K., Marques, J.V., Wendler, J.P., McDermott, J.E., Weitz, K.K., Clauss, T.R., Monroe, M.E., Moore, R.J., Brown, J., Lipton, M.S., Bell, C.J., Pasa Tolic, L., Davin, L.B., and Lewis, N.G. 2021. New insights into lignification via network and multi-omics analyses of arogenate dehydratase knock-out mutants in Arabidopsis thaliana. Front. Plant Sci. 12: 664250. http://dx.doi.org/

Overbey, E.G., Saravia-Butler, A.M., Zhang, Z., Rathi, K.S., Fogle, H., da Silveira, W.A., Barker, R.J., Bass, J.J., Beheshti, A., Berrios, D.C., Blaber, E.A., Čekanaviciute, E., Costa, H.A., Davin, L.B., Fisch, K.M., Gebre, S.G., Geniza, M., Gilbert, R., Gilroy, S., Hardiman, G., Herranz, R., Kidane, Y.H., Kruse, C.P.S., Lee, M.D., Liefeld, T., Lewis, N.G., McDonald, J.T., Meller, R., Mishra, T., Perera, I.Y., Ray, S., Reinsch, S.S., Rosenthal, S.B., Strong, M., Szewczyk, N.J., Tahimic, C.G.T., Taylor, D.M., Vandenbrink, J.P., Villacampa, A., Weging, S., Wolverton, C., Wyatt, S.E., Zea, L., Costes, S.V., and Galazka, J.M. 2021. NASA GeneLab RNA-Seq Consensus Pipeline: Standardized processing of short-read RNA-Seq data. iScience 24: 102361. http://dx.doi.org/

Barker, R.J., Kruse, C.P.S., Johnson, C., Saravia-Butler, A., Fogle, H., Chang, H.-s., Moller Trane, R., Kinscherf, N., Villacampa, A., Manzano, A., Herranz, R., Davin, L.B., Lewis, N.G., Perera, I., Wolverton, C., Gupta, P., Jaiswal, P., Reinsch, S.S., Wyatt, S., and Gilroy, S. 2023. Meta-analysis of the space flight and microgravity response of the Arabidopsis plant transcriptome. npj Microgravity 9: 21. http://dx.doi.org/

Hixson, K.K., Kim, S.J., Moinuddin, S.G.A., Meng, Q., Marques, J.V., Lipton, M.S., Pasa Tolic, L., Davin, L.B., and Lewis, N.G. 2024. Proteomic, metabolomic, and vasculature responses in arogenate dehydratase modulated Arabidopsis. J. Proteome Res. (Manuscript in finalization).

Newsletters:

Rickey T. "A matter of gravity: Understanding how plants grow in space" Molecular Bond, EMSL Newsletter, June 29, 2018. See article at: https://

Seminars/Lectures:

Lewis, N.G., Approach to lignification on Earth and on the International Space Station. Tarim University, Alar, Xinjian, China, June 15, 2018.

Lewis, N.G., Approach to lignification on Earth and on the International Space Station. Southwest University for Nationalities, Chengdu, Sichuan, China, June 24, 2018.

Hanson, DT. Extraterrestrial botany: Fundamental research and inspirational education. A general public interactive lecture at the Science on Tap series, May 2, 2019, O'Neil's, Albuquerque, NM. Hosted by Explora and the National Museum of Nuclear Science and History.

Hanson, DT. Plants in Space! A general public interactive lecture for the Interesting Conversations Series, May 10, 2019, Pentola Restaurant, Albuquerque, NM. Hosted by Sheryl Brown.

Presentations:

Pezeshki, C.C., Costa, M.A., Moinuddin, S.G.A., Davin, L.B., and Lewis, N.G. An integrated omics guided approach to lignification and gravitational responses in International Space Station: The final frontier. WSU Showcase, March 31, 2017.

Lewis, N.G., Davin, L.B., Hanson, D.T., Lipton, M.S., Sayre, R.T. and Starkenburg, S.R. An integrated omics guided approach to lignification and gravitational responses: The final frontier. Joint CSA/ESA/JAXA/NASA Science Symposium for ISS Increments 53&54. Johnson Space Center, Houston TX, July 11-13, 2017.

Pezeshki, C.C., Costa, M.A., Moinuddin, S.G.A., Davin, L.B. and Lewis, N.G. An integrated omics approach to lignification and gravitational responses on ISS: The final frontier. 33rd Annual Meeting American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017.

Lewis, N.G., Costa, M.A., Moinuddin, S.G.A., Davin, L.B., Hanson, D.T., Lipton, M.S., Sayre, R.T. and Starkenburg, S.R. An integrated omics guided approach to lignification and gravitational responses: The final frontier. 33rd Annual Meeting American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017.

Turpin, M., Hanson, D., Lewis, N.G. and Davin, L.B. Gravitational effects on the growth of modified Arabidopsis thaliana in the ISS Advanced Plant Habitat. 34th Annual Meeting American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018.

Lewis, N.G., Costa, M.A., Moinuddin, S.G.A., Davin, L.B., Richards, J., Monje, O., Levine, H., Onate, B., Dufour, N., Carver, J.C., Hanson, D.T., Turpin, M., Lipton, M.S., Hixson, K.K., Starkenburg, S.R. and Sayre, R.T. Arabidopsis GRO (Gravitational Response Omics): Plant Habitat (PH-01). 34th Annual Meeting American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018.

Turpin, M.M., Serda, B.M., Hanson, D.T., Monje, O., Richards, J.T., Carver, J. , Dimapilis, D., Levine, H.G., Dufour, N., Onate, B., Davin, L.B. and Lewis, N.G. Effects of the ISS spaceflight environment and lignin reductions on plant anatomy and gas diffusion in leaves. 35th Annual Meeting American Society for Gravitational and Space Research, Denver, CO, November 20 - 23, 2019 (poster).

Hanson, D.T., Turpin, M.M., Serda, B.M., Monje, O., Richards, J.T., Carver, J. C, Dimapilis, D., Levine, H.G., Dufour, N., Onate, B., Davin, L.B. and Lewis, N.G. Measurements of photosynthesis via pulse-amplitude modulated chlorophyll fluorescence on the ISS. 35th Annual Meeting American Society for Gravitational and Space Research, Denver, CO, November 20 - 23, 2019 (poster).

Monje, O., Richards, J.T., Hanson, D.T., Turpin, M.M., Moinuddin, S.G.A., Costa, M.A., Davin, L.B., Lewis, N.G., Carver, J., Dimapilis, D., Levine, H.G., Dufour, N. and Onate, B. New perspectives for watering substrate-based root modules in microgravity in the Advanced Plant Habitat (APH). 35th Annual Meeting American Society for Gravitational and Space Research, Denver, CO, November 20 - 23, 2019 (oral).

Hixson, K.K., Costa, M.A., Moinuddin, S.G.A., Engbrecht, K.M., Weitz, K.K., Hanschen, E.R., Starkenburg, S.R., Sayre, R.T., Lipton, M.S., Hanson, D.T., Monje, O., Richards, J.T., Davin, L.B. and Lewis, N.G. Investigation of spaceflight environment effects on differentially expressing lignin and carbon capture pathways in Arabidopsis using integrated omics methods. 35th Annual Meeting American Society for Gravitational and Space Research, Denver, CO, November 20 - 23, 2019 (oral).

Hanschen, E.R., Hixson, K.K., Costa, M.A., Moinuddin, S.G.A., Engbrecht, K.M., Weitz, K.K., Starkenburg, S.R., Sayre, R.T., Lipton, M.S., Hanson, D.T., Monje, O., Richards, J.T., Davin, L.B. and Lewis, N.G. Patterns of gene expression in Arabidopsis thaliana in response to microgravitational environment. 35th Annual Meeting American Society for Gravitational and Space Research, Denver, CO, November 20 - 23, 2019 (oral).

Lewis, N.G. Multi-omics analyses of ISS grown lignin-reduced Arabidopsis lines: New insights into potential of genetically modified vascular plants in the space environment. 37th Annual Meeting American Society for Gravitational and Space Research, Baltimore, MD, November 3 - 6, 2021.

Lewis, N.G. Multi-omics analyses of lignin reduced Arabidopsis lines on ISS. NASA's Associate Administrator of the Science Mission Directorate (SMD), Washington DC, December 2, 2021.

Lewis, N.G. Integrated multi-omics analyses of lignin-reduced Arabidopsis lines on International Space Station. 61st Annual Meeting, Phytochemical Society of North America, Virginia Tech, Blacksburg, VA, July 24 - 28, 2022.

Heinse, R., Monje, O., Richards, J., Dufour, N. and Lewis, N.G. Root-zone water stress assessment for the Advanced Plant Habitat (APH) inaugural mission on the International Space Station (ISS). COSPAR 22, 44th Scientific Assembly, Athens, Greece, July 16 - 24, 2022.

Lewis, N.G., Davin, L.B., Costa, M.A., Moinuddin, S.G.A., Hanschen, E.R., Starkenburg, S.R., Koehler, S.I., Winnacott, B.J., Hixson, K.K., Lipton, M.S., Hanson, D.T., Turpin, M.M., Monje, O., Richards, J.T., Dufour, N. and Levine, H.G. Integrated multi-omics analyses of lignin-reduced Arabidopsis lines on International Space Station. 38th Annual Meeting American Society for Gravitational and Space Research, Houston, TX, November 9 - 11, 2022.

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Media Coverage:

• Watson KE (Staff Writer). "McCaffrey students visited by space explorer: Astronaut Sunita Williams shares her experiences." The Galt Herald, April 18 2018. See article at http://www.galtheraldonline.com/

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• Feustel AJ (Drew) "VIDEO from space, harvest Day! Farmer's almanac update on International Space Station. The plants continue to grow under the watchful eye of #AstroFarmer @astro_ricky! Twitter. July 24, 2018. Listen at https://

• Fans of science in space now can experience fast-moving footage in even higher definition as NASA and ESA (European Space Agency) deliver the first 8K ultra high definition (UHD) video of astronauts living, working and conducting research from the International Space Station. https://

• "Experience space station research in high res video". WSU Insider, November 9, 2018. See article at https://

• "NASA Space Biology Shared a Post," Facebook, Internet. (February 1, 2019). Success! From splashdown of SpaceX16's Dragon capsule to Dr. Norman Lewis' lab. Our #PlantHabitat01 samples arrived safely at Washington State University. Ground controls arrived from

Kennedy Space Center, analysis will soon begin! Stay tuned for another peek behind the scenes...See post at: https://

• Washington Grown, Season 6, Episode 13: This episode is out of this world! Not only will we be talking about herbs, including visiting Steel Wheel Farm and Mesa de Vida, but we'll also head to Washington State University to learn about the Final Frontier Plant Habitat - where we talk about growing plants in space! Original Air Date: 4/27/19. See http://www.wagrown.com/

Training:

Miss Bianca Serda, a Hispanic student with Dr. Hanson gave three oral and three poster presentations. She was the recipient of a NASA Space Life Science Training Program (SLSTP) fellowship at Ames Research Center (from June 10, 2019, to August 16, 2019), results of which she presented at the ASGSR meeting in Denver.

• Oral Presentations:

o Serda, B., Turpin, M., Hudson, P., and Hanson, D. “Plants in Space: Interactions between Morphology, Lignification, and Carbon Isotopic Composition.” UNM Hispanic Youth and Leadership Program, Albuquerque, NM, November 2019.

o Serda, B., Turpin, M., Hudson, P., and Hanson, D. “Plants in Space: Interactions between Morphology, Lignification, and Carbon Isotopic Composition.” UNM Center for Stable Isotopes Seminar, Albuquerque, NM, October 2019

o Serda, B., Turpin, M., Lewis, N.G., and Hanson, D. “Plants in Space: Do Lignification Levels and Microgravity Interact to Impact Photosynthesis? University of New Mexico Department of Biology 28th Annual Research Day, Albuquerque, NM, March 2019.

• Poster Presentations:

o Serda B, Turpin M, Hudson P, Hanson D. “Plants in Space: Interactions between Morphology, Lignification, and Carbon Isotopic Composition.” SACNAS Conference at San Antonio, TX. October 2018.

o Serda B, Turpin M, Hudson P, Hanson D. “Plants in Space: Interactions between Morphology, Lignification, and Carbon Isotopic Composition.” Presented at the MARC Symposium, Albuquerque, NM, August 2018.

o Serda B, Turpin M, Hudson P, Hanson D. “Plants in Space: Interactions between Morphology, Lignification, and Carbon Isotopic Composition.” New Mexico Academy of Science Research Symposium at Albuquerque, NM. October 2018.

o Serda, B., Turpin, M., Hudson, P., and Hanson, D. “Plants in Space: Interactions between Morphology, Lignification, and Carbon Isotopic Composition.” SACNAS Conference, Honolulu, HI, November 2019.

o Serda, B., Turpin, M., Hudson, P., and Hanson, D. “Plants in Space: Interactions between Morphology, Lignification, and Carbon Isotopic Composition.” New Mexico AMP Conference, New Mexico State University, Albuquerque, NM, October 2019.

Pullman High School student (PHS Senior Charles Pezeshki) was involved in the WSU project. He carried out/refined Arabidopsis growth conditions using the Science Carrier that was to be fitted into the APH. The Washington State Space Grant Consortium kindly provided supplementary support for both high school and undergraduate participation. Mr. Pezeshki successfully completed his Senior Project, and which led to several successful outcomes. He gave two posters (one at WSU for the University Showcase, and a second one in October 2017 in Seattle at the Annual Meeting of the American Society for Gravitational and Space Research (ASGSR) in a session for High School students nationwide). He was also supported by a travel grant from ASGSR to present his work. Some of Mr. Pezeshki’s activities included:

• Sieving, washing, and autoclaving of Turface arcillite particles used as the growing substrate for plants on board ISS, as well as preparation of foam and gauze templates to support the seeds and plants in the final assembly of the Science Carrier unit.

• Recording and editing videos to present to NASA KSC personnel, so that they could review our plant harvesting and processing methods used for RNA, protein, and metabolite extraction. These videos were instrumental for assessing constraints involved in foil pack dimensions required for plant samples for adequate preservation during storage and for rapid processing upon return to Earth. This was helpful for developing procedures necessary for optimizing astronaut crew time spent on harvesting and critical storage of samples in the limited freezer container space aboard ISS.

• Assistance with transgenic plant sample preparation for PCR gene screening analysis. This involved DNA extraction from freshly harvested plant leaves, and use of these samples as DNA templates for the PCR mixtures. Samples were amplified using a thermocycler instrument and then run on an agarose gel for a qualitative analysis assessment.

• Developing protocols for harvesting and processing different parts of Arabidopsis plants which included separating leaves, stems, siliques, flowers, etc., and then later on grinding tissues and preparing extracts suitable for our multi-omics studies. He continuously assisted in preparing plant tissue samples for metabolomic analysis. He also prepared a video presentation on this that was also very helpful for the NASA scientists/astronauts by recording this entire process beginning from harvest to extraction.

• Studying effects of humidity on metabolite levels at different harvest times. Mr. Pezeshki individually processed leaf and stem tissues of four and six weeks old Arabidopsis wild type plants grown under low and high humidity conditions. He studied the growth and development (stem length and weight) of Arabidopsis wild type plants harvested at different time intervals. He also successfully completed this experiment without assistance. After he had expertise in preparing tissue samples suitable for metabolomics, he was assigned to study the metabolomics of NASA-SVT (Science verification test) samples.

PhD students Kim K. Hixson, Qingyan Meng and Jorgen Tacetta: Both K.K. Hixson and Q. Meng completed their PhD degrees (N. G. Lewis, Advisor) in the Molecular Plant Sciences program, Washington State University. Dr. Hixson is currently a Senior Research Scientist at PNNL, and Dr. Meng is a Research Associate in the Lewis Laboratory). Jorgen Tacetta is in good standing in his Ph.D. studies.

Outreach:

• The New Mexico Space Grant Consortium helped to fund a year-long collaboration between Dr. Hanson and the Explora science museum in Albuquerque. They developed and piloted a monthly course called "Extraterrestrial Botany" for around twenty students across the city who were in 5th through 9th grade. Ten sessions were developed and taught by a team of 2 post docs (1 female), two technicians (both female, 1 Hispanic), a female graduate student, and three undergraduates (2 female, 1 Hispanic), plus Dr. Hanson and two staff members at Explora. The project involved learning about challenges of space exploration, relevance for life on Earth, and the student assisted with data collection and analysis from the APH-01. Feedback from families and Explora was extremely positive.

• David Hanson was lead organizer for a NASA supported In-flight Education Downlink with Astronaut Christina Koch on International Space Station (https:// This event had an audience of over 700 people, including over 500 K-12 students that traveled to UNM to participate in person - as well as others watching via an internet live stream (live.unm.edu). The event had many hands-on activity booths from local STEM education programs and guest presentations by NASA scientists and student interns. It was also co-hosted by The Children’s Hour radio program and part of a subsequent hour-long show that was broadcast nationwide. Research from PH-01 was highlighted in the live event (found online at live.unm.edu as well as directly through the NASA past downlinks site- https:// and on the radio show (https://

• Norman G. Lewis and Laurence B. Davin partnered with Ms. Kathy Lucchesi (K-7/8 teacher), at McCaffrey Middle School in Galt, California, and their largely Hispanic students. Supplemental funding was provided to the school by NASA and the California Space Grant Consortium so that these middle school students could safely follow and utilize many of the plant growth and development protocols developed for the ISS experiments. One purpose was that the students could grow plants under similar conditions and obtain information and insights on how the research impacts or benefits life on Earth and beyond (in the future).

o Drs. Lewis (Washington State University) and Dr. Sato (NASA Ames) visited McCaffrey Middle School on September 28, 2017, to see first-hand the work underway. Both visitors gave talks on the project, as well as the broader ramifications of NASA research.

o The Galt Herald wrote an article on this visit (http://www.galtheraldonline.com/ and the student’s progress was further publicized through the NASA Space Biology Facebook (https://

o Progress with the students was also followed through regular Skype and/or FaceTime team meetings, with Drs. Lewis, Davin, and Costa (WSU). Dr. Hanson (University of New Mexico) provided tutorials on photosynthesis and the use of a FluorPen for their studies.

o Written materials on, and seeds for, the experiments at hand were also routinely provided. The additional aim was in helping teach and inspire these young students about the joys and fun of the scientific method in experimental plant biology. Periodically, middle school students presented results to Dr. Lewis over where such work was being routinely evaluated.

o Astronaut Sunita Williams, Dr. Lewis (WSU IBC) and Dr. Sato (NASA Ames) visited the school on April 13, 2018. Astronaut Williams gave presentations on her time on International Space Station and answered questions from students. The Galt Herald (http://www.galtheraldonline.com/ and the Lodi News-Sentinel (www.lodinews.com/news/article_982e5434-3faa-11e8-b4e1-db589601e0df.html?mode=story) wrote articles on her visit, and the visit was further publicized through the NASA Space Biology Facebook (https://

o McCaffrey Middle School Students also presented their work on #PlantHabitat01 to the Galt School Board on (See post at: https://

o Ms. Lucchesi, her students, and Dr. Lewis visited NASA AMES Research Center, May 28, 2018. The students presented their research to NASA scientists, engineers, and managers. The students also handled small satellites, checked out NASA’s bone and signaling lab, and learned about Astrobiology and Air Revitalization on ISS. (See post at: https://

o Ms. Lucchesi and one of her students traveled to Washington State University, Pullman to visit Dr. Lewis’ and Davin’s laboratory. They received hands-on training on what was being done with the Arabidopsis plants. (See post at: https://

o McCaffrey Middle School students were featured on a Good Day Sacramento segment between 8-10 AM, Thursday, October 25, 2018! (See Part 1: https:// and Part 2: https:// with the Lewis lab personnel included from WSU (via teleconference).

• On November 7, 2018, and January 3, 2019, the Pullman Droids, a Lego Robotics Team, visited Drs. Lewis and Davin to learn about the PH-01 ISS study.

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